

## SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

### Sub-Strand 2.4: Area of Polygons — Teacher Reference (Pre-filled)

| SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10 |                                                                                                                                                                                                                                       |
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| Sub-Strand                                    | Sub-Strand 2.4: Area of Polygons                                                                                                                                                                                                      |
| Driving Question                              | DRIVING QUESTION: How can we calculate the exact area of any polygon-shaped surface — even when we cannot measure its height — so that builders, farmers, and designers in our community never order too much or too little material? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                                                                                                  | What did I observe?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | What did I learn?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>Lesson 1</b><br><b>Lesson 1: Anchoring the Driving Question — Area of Triangles Using <math>\frac{1}{2}ab \sin C</math></b>                      | <p>Teacher presents the anchoring phenomenon: a Kericho tea-estate manager needs to tile an oddly shaped triangular courtyard, but the perpendicular height cannot be measured because a concrete pillar is in the way. Students observe that standard base <math>\times</math> height formulas fail when height is physically inaccessible. Teacher then guides students to measure (or read off) two sides and the included angle of scale-diagram triangles cut from manila paper, establishing the need for a new tool. Pedagogical note: Use cold-calling (minimum 4 students) during the predict phase — ask 'What information do you think is enough to fix a triangle's size?' before any instruction is given. Allow 30-second wait time after posing the question.</p> | <p>Students learn that the area of any triangle can be calculated using two sides and the included angle via the formula <math>\text{Area} = \frac{1}{2}ab \sin C</math>. They establish that <math>\sin C</math> acts as a 'height-without-measuring' multiplier: the perpendicular height <math>h = b \sin C</math>, so <math>\text{Area} = \frac{1}{2}a(b \sin C) = \frac{1}{2}ab \sin C</math>. Key vocabulary secured: included angle, opposite side, perpendicular height. Pedagogical note: Insist students write the formula in words first ('half the product of two sides times the sine of the angle between them') before writing symbols, to anchor meaning.</p> | <p>Students explain why the formula works by annotating a triangle diagram: they draw the altitude from vertex C to side c, label it <math>h = b \sin C</math>, and substitute into the familiar <math>\frac{1}{2} \times \text{base} \times \text{height}</math>. The formula is then applied to the Kericho courtyard triangle (e.g., <math>a = 14 \text{ m}</math>, <math>b = 18 \text{ m}</math>, <math>C = 72^\circ</math>) to calculate the exact tile area needed. Teacher asks: 'Would the answer change if we used a different pair of sides and their included angle? Why or why not?' — directing students to see that all three versions (<math>\frac{1}{2}ab \sin C</math>, <math>\frac{1}{2}bc \sin A</math>, <math>\frac{1}{2}ac \sin B</math>) are equivalent. Pedagogical note: Display all three equivalent forms side-by-side on the board; require students to verify numerically with one worked example.</p> |
| <b>Lesson 2</b><br><b>Lesson 2: Triangle Area Without a Height — <math>\frac{1}{2}ab \sin C</math> in Practice &amp; Heron's Formula Introduced</b> | <p>Teacher poses a new constraint on the Kericho courtyard problem: the estate surveyor has measured only the three side lengths of a triangular plot — no angles were recorded. Students observe that <math>\frac{1}{2}ab \sin C</math> cannot be applied directly because no</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Students learn Heron's Formula: <math>\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}</math>, where <math>s = (a+b+c)/2</math> is the semi-perimeter. They learn that <math>s</math> is computed first, then each factor (<math>s-a</math>), (<math>s-b</math>), (<math>s-c</math>) is evaluated and the product taken under the square root. Key insight:</p>                                                                                                                                                                                                                                                                                                                       | <p>Students apply Heron's Formula to the Kericho courtyard triangle and verify the result matches the <math>\frac{1}{2}ab \sin C</math> answer from Lesson 1 (for the 9-12-15 right triangle, both methods yield <math>54 \text{ m}^2</math>). Teacher then assigns a non-right triangle (e.g., a Nairobi plot with</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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|                                                                                                                                            | <p>angle value is known. Teacher displays the data set (e.g., <math>a = 9</math> m, <math>b = 12</math> m, <math>c = 15</math> m) on the board and asks students to predict whether area can still be found exactly. Students observe a teacher-guided derivation on the board showing how the cosine rule can extract angle C from side lengths, and how that value feeds back into <math>\frac{1}{2}ab \sin C</math>, producing — after algebraic manipulation — Heron's Formula.</p> <p>Pedagogical note: Pause the derivation at the step '<math>\cos C = (a^2+b^2-c^2)/2ab</math>' and cold-call 3 students to name the rule being used before proceeding.</p>                                              | <p>Heron's Formula is an all-sides formula — it requires no angle measurement and no altitude measurement, making it ideal for surveyed land parcels. Pedagogical note: Require students to write <math>s</math> and all three bracket values explicitly before multiplying — do not allow skipped steps in classwork.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>sides 7 m, 11 m, 13 m) so students see Heron's Formula working where <math>\frac{1}{2}</math> base <math>\times</math> height would require extra steps.</p> <p>Discussion question: 'A farmer in Kisumu knows only the three fence-post distances around a triangular shamba. Which formula should she use and why?' — expected answer: Heron's Formula, because no height or angle data is available.</p> <p>Pedagogical note: Circulate during pair work; listen for students confusing perimeter <math>s</math> with semi-perimeter — correct immediately.</p>                                                                                                                                                                                         |
| <p><b>Lesson 3</b><br/><b>Lesson 3: Deep Practice &amp; Model Building — Heron's Formula Consolidated</b></p>                              | <p>Teacher opens with three worked examples on the board ranging in difficulty: (i) a scalene triangle with integer sides; (ii) a triangle where one <math>(s-x)</math> factor is a decimal; (iii) a real-world Rift Valley farm plot where students must first convert units (acres to <math>m^2</math>) before applying the formula. Students observe that the formula's structure is symmetric — swapping labels <math>a</math>, <math>b</math>, <math>c</math> does not change the product <math>s(s-a)(s-b)(s-c)</math>.</p> <p>Pedagogical note: After Example (i), pause and ask 4 cold-called students to predict the answer for Example (ii) before revealing working — allow 45-second think time.</p> | <p>Students consolidate the procedural fluency of Heron's Formula and learn its symmetry property: the area result is independent of how the three sides are labelled. They also learn to identify when Heron's Formula is more efficient than <math>\frac{1}{2}ab \sin C</math> (when only side lengths are given) versus when <math>\frac{1}{2}ab \sin C</math> is more efficient (when two sides and included angle are given). Pedagogical note: Post a decision flowchart on the board — 'Do you know an angle? → Yes: use <math>\frac{1}{2}ab \sin C</math>. No: use Heron's.' Students copy this into their exercise books as a reference tool.</p>                                                                                                                     | <p>Students explain their reasoning in writing: given a Mombasa architect's triangular garden with sides 8 m, 15 m, 17 m, they must (a) apply Heron's Formula, (b) state which method they would use if they also knew the angle between the 8 m and 15 m sides, and (c) justify why both answers must be identical. Model-building phase: students update their cumulative courtyard diagram, annotating the triangular section with the correct formula and computed area. Pedagogical note: Collect and scan written justifications — common error to address in Lesson 4 is premature rounding of <math>s</math> before computing bracket values.</p>                                                                                                     |
| <p><b>Lesson 4</b><br/><b>Lesson 4: Area of Rhombus and Parallelogram — Measuring the Stage Floor Without Dropping a Perpendicular</b></p> | <p>Teacher displays a photograph of a diamond-patterned Maasai bead-work panel and a parallelogram-shaped stage floor in a Nairobi school hall. Students observe that both shapes have no easily accessible right angle — a perpendicular height would require scaffolding. Teacher demonstrates, using two metre rulers as diagonals, that the diagonals of a rhombus bisect each other at right angles, creating four congruent right triangles. Students observe the diagonal-bisection property physically before any formula is stated.</p> <p>Pedagogical note: Have two students hold the 'diagonals' while the class counts the four right triangles — kinesthetic anchoring is important here.</p>      | <p>Students learn two formulas: (1) Rhombus: <math>A = \frac{1}{2}d_1d_2</math>, derived by summing the four right triangles each of area <math>\frac{1}{2}(d_1/2)(d_2/2)</math>. (2) Parallelogram: <math>A = ab \sin \theta</math>, derived by recognising that the parallelogram is two congruent triangles, each of area <math>\frac{1}{2}ab \sin \theta</math>, giving total area = <math>ab \sin \theta</math>. Key distinction: for the rhombus, <math>d_1</math> and <math>d_2</math> are diagonal lengths; for the parallelogram, <math>a</math> and <math>b</math> are adjacent side lengths and <math>\theta</math> is the included angle. Pedagogical note: Students frequently confuse 'side' and 'diagonal' — insist on labelled diagrams for every problem.</p> | <p>Students apply both formulas to the Kericho courtyard model: a rhombus-shaped flower bed (<math>d_1 = 6</math> m, <math>d_2 = 8</math> m → <math>A = 24</math> <math>m^2</math>) and a parallelogram-shaped paving section (<math>a = 5</math> m, <math>b = 7</math> m, <math>\theta = 65^\circ</math> → <math>A \approx 31.84</math> <math>m^2</math>).</p> <p>Teacher asks: 'Why do architects prefer these formulas over measuring perpendicular height on a building site?' — expected answer: diagonals and angles are easier to measure with a tape and clinometer. Pedagogical note: Require students to state units in every answer; deduct marks during marking if <math>m^2</math> is omitted — this builds scientific communication habits.</p> |

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| <p><b>Lesson 5</b><br/><b>Lesson 5: Practise &amp; Apply — Rhombus and Parallelogram in Context</b></p>                                  | <p>Teacher presents a multi-context practice carousel: Station A — a Kisumu tiler calculating rhombus-shaped floor tiles (diagonals given); Station B — a Rift Valley farmer fencing a parallelogram plot (sides and angle given); Station C — a Nairobi graphic designer scaling a parallelogram logo (proportional sides, angle unchanged). Students rotate through stations in groups of 4, recording observations about which formula applies and why. Pedagogical note: Station C deliberately keeps angle <math>\theta</math> constant while doubling side lengths — use this to provoke the question 'What happens to area when sides double?' Answer: area quadruples (scales as <math>s^2</math>). Allow 8 minutes per station.</p>                                                                                                      | <p>Students learn that (i) Area = <math>ab \sin \theta</math> is the practical choice when perpendicular height cannot be physically measured; (ii) Area = <math>\frac{1}{2}d_1d_2</math> requires only a tape measure stretched corner-to-corner — no angle instrument needed; (iii) when both diagonals and an angle are known for a rhombus, either formula should yield the same answer — this serves as a self-checking tool. Students also learn that area scales as the square of the linear scale factor. Pedagogical note: Highlight the self-check strategy explicitly — students who get different answers using both methods have made an arithmetic error somewhere.</p>                                                                         | <p>Students write a structured justification: given a parallelogram paving slab (<math>a = 3</math> m, <math>b = 4</math> m, <math>\theta = 50^\circ</math>), they compute area, then explain in one paragraph why a builder in Mombasa would find this formula more practical than measuring and dropping a perpendicular on a laid concrete surface. Teacher cold-calls 3 students to read their paragraphs aloud, then facilitates a class discussion connecting back to the driving question. DQB update: students add 'rhombus' and 'parallelogram' entries to the class driving-question board with their formulas and real-world use cases.</p>                                                                                       |
| <p><b>Lesson 6</b><br/><b>Lesson 6: Area of a Regular Pentagon — Deriving and Applying <math>A = (5s^2)/(4 \tan 36^\circ)</math></b></p> | <p>Teacher shows a photograph of a Nairobi housing-estate courtyard paved with regular pentagonal stones and asks: 'How did the supplier know how much concrete to mix?' Students observe that a regular pentagon cannot be split into rectangles easily. Teacher then demonstrates — using a pre-drawn large pentagon on chart paper — that it can be divided into 5 congruent isosceles triangles from the centre. Students observe the teacher bisecting one triangle to form a right triangle, labelling the apothem, the half-base (<math>s/2</math>), and the apex half-angle (<math>36^\circ</math>). Pedagogical note: Cold-call 2 students to state the sum of interior angles of a pentagon (<math>540^\circ</math>) and the size of each central angle (<math>72^\circ</math>) before proceeding — this activates prior knowledge.</p> | <p>Students learn that the apothem <math>a = (s/2)/\tan 36^\circ</math>, because the apothem is the adjacent side and <math>s/2</math> is the opposite side in the right triangle formed by bisecting the central isosceles triangle. The area of one triangle = <math>\frac{1}{2} \times s \times (s/2)/\tan 36^\circ = s^2/(4 \tan 36^\circ)</math>. Multiplying by 5 gives the full formula: <math>A = 5s^2/(4 \tan 36^\circ)</math>. Students also learn the decimal approximation <math>\tan 36^\circ \approx 0.7265</math>, giving <math>A \approx 1.720s^2</math>. Pedagogical note: Require students to derive — not just memorise — the formula; derivation should appear in their exercise books before any application problems are attempted.</p> | <p>Students apply <math>A = 5s^2/(4 \tan 36^\circ)</math> to: (i) the Nairobi courtyard pentagonal stone (side = <math>0.4</math> m <math>\rightarrow A \approx 0.275</math> m<sup>2</sup>); (ii) a larger pentagonal classroom display board (side = <math>1.2</math> m). Teacher asks: 'If a designer doubles the side length of the pentagon, by what factor does the area increase?' — expected answer: factor of 4 (area <math>\propto s^2</math>). Students update their cumulative courtyard model by adding a pentagonal feature and labelling it with the formula and computed value. Pedagogical note: Watch for calculator errors with <math>\tan 36^\circ</math> — remind students to ensure calculators are in degree mode.</p> |
| <p><b>Lesson 7</b><br/><b>Lesson 7: Area of a Regular Hexagon — Deriving the Formula from Equilateral Triangles</b></p>                  | <p>Teacher displays a honeycomb image (referencing Kenya's honey industry in Baringo County) and a photograph of hexagonal paving tiles outside a Kisumu government building. Students observe that the hexagon is composed of 6 equilateral triangles — teacher demonstrates this by cutting a paper hexagon and rearranging the triangles. Students observe that each equilateral triangle has side length equal to the hexagon's side length <math>s</math>. Pedagogical</p>                                                                                                                                                                                                                                                                                                                                                                   | <p>Students learn that the area of one equilateral triangle with side <math>s</math> is <math>A = (\sqrt{3}/4)s^2</math>. Multiplying by 6 gives the hexagon formula: <math>A = (3\sqrt{3}/2)s^2</math>. Students also learn the decimal approximation <math>(3\sqrt{3}/2) \approx 2.598</math>, so <math>A \approx 2.598s^2</math>. Key conceptual link: the regular hexagon formula is a direct scaling of the equilateral triangle formula — no new trigonometric derivation is needed. Students compare this to the pentagon derivation and note that the hexagon's formula is 'cleaner'</p>                                                                                                                                                              | <p>Students apply <math>A = (3\sqrt{3}/2)s^2</math> to: (i) a Baringo honeycomb cell (<math>s = 3</math> mm <math>\rightarrow A \approx 23.38</math> mm<sup>2</sup>); (ii) a hexagonal paving tile outside a Kisumu building (<math>s = 0.25</math> m <math>\rightarrow A \approx 0.162</math> m<sup>2</sup>). Teacher asks: 'A hardware shop in Nakuru sells hexagonal tiles at Ksh 850 per m<sup>2</sup>. How much would it cost to tile a hexagonal stage of side 2 m?' Students must compute area then multiply by unit cost. Pedagogical note: This monetised real-world extension connects mathematics</p>                                                                                                                             |

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|                                                                                                                       | note: Ask students to predict the area formula before derivation — accept all predictions and record them on the board; revisit at the end of the lesson to affirm or correct. Allow 1-minute silent think time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | because all interior triangles are equilateral (all angles $60^\circ$ ), whereas the pentagon's triangles are only isosceles. Pedagogical note: This comparison deepens understanding of why regular polygons with more sides are harder to derive formulas for.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | to financial literacy (a PCIs requirement). DQB update: students record the hexagon formula alongside the pentagon formula and note similarities and differences.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Lesson 8</b><br><b>Lesson 8: Synthesis, Model Building &amp; Final Assessment — The Complete Kericho Courtyard</b> | Teacher reveals the complete Kericho courtyard design: a composite polygon made up of a triangular entrance (sides only given — Heron's Formula required), a parallelogram paving section (sides and angle given), a rhombus flower bed (diagonals given), a pentagonal fountain base (side given), and a hexagonal seating area (side given). Students observe the full design for 2 minutes silently, then in pairs annotate which formula applies to each section. Pedagogical note: This is the summative sense-making phase — resist giving hints. Cold-call 5 students to state their formula choice and justification before any whole-class work begins. Record all five on the board. | Students synthesise all five area-of-polygon formulas from the unit: (a) $\text{Area} = \frac{1}{2}ab \sin C$ — triangles with two sides and an included angle; (b) Heron's Formula — triangles with three sides only; (c) $A = \frac{1}{2}d_1d_2$ — rhombus with both diagonals; (d) $A = ab \sin \theta$ — parallelogram with adjacent sides and included angle; (e) $A = 5s^2/(4 \tan 36^\circ)$ — regular pentagon; (f) $A = (3\sqrt{3}/2)s^2$ — regular hexagon. Students learn that the key skill is formula selection: matching the given information to the correct formula is as important as executing the calculation. Pedagogical note: The formula-selection decision tree (introduced in Lesson 3 and expanded each lesson) should now be fully populated in students' notes — use the last 5 minutes of the lesson to have students check and complete it. | Students independently calculate the total area of the Kericho courtyard by summing all five section areas, then answer: 'The contractor charges Ksh 2,400 per $\text{m}^2$ for tiling. What is the total cost? If the budget is Ksh 500,000, will the project be within budget?' This integrates arithmetic, formula application, unit consistency, and financial reasoning. Final model-building phase: students produce a fully annotated scale diagram of the courtyard with every formula, substituted values, and computed area clearly labelled — this serves as the unit's summative portfolio artefact. Pedagogical note: Mark using a two-column rubric — one column for mathematical accuracy (formula, substitution, computation) and one for communication (labelling, units, justification). Return marked work with written feedback before the next sub-strand begins. |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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